

Response to Reviewers

Manuscript PONE-D-26-28965

Evaluation of validation strategies and transferability of EEG attention decoding: a comparative analysis on open datasets

[TODO — working state of this letter; this banner is not part of the submission.] Every response below is in one of three states. A response with no marker is backed both by the code in `python_project/` and by text that is actually present in `manuscript_clean/manuscript.tex`. A response followed by a **[TODO]** paragraph is only partly backed: the implementation exists, but the manuscript does not yet say so, or the numbers it refers to have not been computed. A response consisting only of **[TODO]** is not answered at all yet. The manuscript is mid-rewrite — its Abstract, Background, Models and Results sections are cleared placeholders — the full sweep in `python_project/results/` is still running, and no analysis has been produced from it. Until that changes, no response here may cite a number, a table, a figure or a manuscript passage. All markers must be gone before the letter is uploaded.

Dear Editor and Reviewers,

We thank the Editor and the Reviewers for their careful evaluation of our manuscript and for their constructive comments. We have carefully considered all comments and are preparing a revised version of the manuscript that addresses the issues raised during peer review.

The major revisions will include:

- **[TODO — major revision 1: redefinition of the target construct.]** The direction is settled — the study claims discrimination between operationally defined experimental conditions rather than decoding of attention as a psychological construct — but the manuscript does not yet carry it: Table 1 still names its classes “High Attention” and “Low Attention”, and no label-mapping sensitivity analysis exists in the code. Only the exclusion of drowsy epochs from the primary analysis is in place. Rewrite this bullet together with the responses to MR1 and Comments 3.1, 3.2 and 4.5.
- *[Major revision 2 — to be completed after the corresponding manuscript changes are finalized.]*
- *[Major revision 3 — to be completed after the corresponding manuscript changes are finalized.]*
- *[Additional major revision, if needed.]*

For clarity, our response is organized into four sections:

1. **Journal Requirements**
2. **Reviewer #1**
3. **Reviewer #2**
4. **Reviewer #3**

Reviewer and journal comments are reproduced below and followed by our point-by-point responses. Where a single original paragraph contains several independent criticisms or requests, it is divided into separate response points so that each issue can be addressed explicitly. Changes to the manuscript are identified with the corresponding page and line numbers wherever applicable.

Yours sincerely,

[TODO: author list]

Journal Requirements

The seven requirements listed in the decision letter are addressed below.

Comment 1

Please ensure that your manuscript meets PLOS ONE's style requirements, including those for file naming. The PLOS ONE style templates can be found at https://journals.plos.org/plosone/s/file?id=wjVg/PLOOne_formatting_sample_main_body.pdf and https://journals.plos.org/plosone/s/file?id=ba62/PLOOne_formatting_sample_title_authors_affiliations.pdf

Response:

[TODO]

Comment 2

Please note that PLOS One has specific guidelines on code sharing for submissions in which author-generated code underpins the findings in the manuscript. In these cases, we expect all author-generated code to be made available without restrictions upon publication of the work. Please review our guidelines at <https://journals.plos.org/plosone/s/materials-and-software-sharing#loc-sharing-code> and ensure that your code is shared in a way that follows best practice and facilitates reproducibility and reuse.

Response:

[TODO]

Comment 3

We note that the grant information you provided in the 'Funding Information' and 'Financial Disclosure' sections do not match. When you resubmit, please ensure that you provide the correct grant numbers for the awards you received for your study in the 'Funding Information' section.

Response:

[TODO]

Comment 4

Thank you for stating the following financial disclosure:

"The author(s) declared that financial support was received for this work and/or its publication. This research was funded by the Committee of Science of the Ministry of Science and Higher Education of the Republic of Kazakhstan (Grant no. BR27198099)."

Please state what role the funders took in the study. If the funders had no role, please state: "The funders had no role in study design, data collection and analysis, decision to publish, or preparation of the manuscript."

If this statement is not correct you must amend it as needed.

Please include this amended Role of Funder statement in your cover letter; we will change the online submission form on your behalf.

Response:

[TODO]

Comment 5

Please note that your Data Availability Statement is currently missing the repository name and/or the DOI/accession number of each dataset OR a direct link to access each database. If your manuscript is accepted for publication, you will be asked to provide these details on a very short timeline. We therefore suggest that you provide this information now, though we will not hold up the peer review process if you are unable.

Response:

[TODO]

Comment 6

We notice that your supplementary figures are uploaded with the file type 'Figure'. Please amend the file type to 'Supporting Information'. Please ensure that each Supporting Information file has a legend listed in the manuscript after the references list.

Response:

[TODO]

Comment 7

If the reviewer comments include a recommendation to cite specific previously published works, please review and evaluate these publications to determine whether they are relevant and should be cited. There is no requirement to cite these works unless the editor has indicated otherwise.

Response:

[TODO]

Reviewer #1

Comment 1

MR1. The binary attention labels have uncertain construct validity. Dataset B was collected for stress analysis rather than attention measurement, while Dataset A uses instructed states presented in a fixed temporal sequence. Drowsiness also involves eye closure. Consequently, the classes confound attention with stress, arousal, visual input, ocular activity and time-on-task.

Response:

We thank the Reviewer for identifying this fundamental issue. We agree that the original framing overstated what the available labels can support: they are not independent measurements of an individual’s attention, and differences in stress, arousal, sensory input, ocular activity, time-on-task, and other protocol features may contribute to discriminability.

We therefore revised the research question, motivation, and knowledge gap throughout the manuscript. The study no longer treats its results as evidence for or against invariant neural markers of attention. Instead, it evaluates zero-shot transfer of a *protocol-defined* contrast between higher-demand conditions and rest or task disengagement, and reports pre-adaptation benchmarks for source–target suitability and negative transfer. The revised Introduction states this boundary of inference explicitly. We also excluded the *Drowsy* phase because eye closure and drowsiness would add pronounced differences in arousal, visual input, and ocular activity. The remaining confounds are acknowledged as limitations rather than interpreted as attention effects.

Comment 2

MR2. The principal within-dataset baseline is affected by substantial information leakage. One-second windows separated by only 0.125 seconds share 87.5% of their samples, yet temporally adjacent windows are distributed across stratified folds. Acknowledging this issue as a limitation does not make the reported baseline scientifically valid.

Response:

We thank the Reviewer for drawing our attention to this issue. The original random split did not adequately account for the dependence between densely overlapping windows.

We reconstructed the baseline using group-aware splitting at the recording-unit level: all windows from the same recording unit are assigned wholly to either the training or the test partition. Thus, no temporally adjacent overlapping windows are split between partitions. We also replaced the previous 1-s windows with 5-s windows and 0.5-s overlap, and recalculated all reported results under this leakage-resistant design.

Comment 3

MR3. Several validation procedures are insufficiently defined. Dataset B trials appear to be treated as sessions without justification; the composition of the “task-only” datasets is unclear; class counts are absent; and cross-task testing does not separate task transfer from participant identity.

Response:

We thank the Reviewer for this clarification. We added a separate cross-trial strategy for SAM-40. We initially treated its separately provided task recordings as session-like, but the source publication describes the three repetitions as trials and does not establish separate physical sessions; cross-session validation therefore now applies only to Dataset A.

Dataset compositions and class counts. We added Table ??, which defines the full, task-only, and two-task Dataset B compositions by their retained conditions. It also reports the number of participants, recording units, and prepared pre-oversampling windows in the low- and high-attention classes for each composition.

Subject-identity confounding in cross-task testing. We rebuilt cross-task validation as leave-one-subject-out transfer: for each fold, the source-task training data exclude subject S , and the target-task test data contain only subject S . Source and target partitions are therefore subject-disjoint by construction.

Comment 4

MR4. Reproducibility is incomplete within the manuscript. Filter type and order, ICA implementation and rejection criteria, STFT window and FFT parameters, feature dimensionality, scaler-fitting scope, XGBoost hyperparameters, tuning procedure, random seeds and fold construction are not adequately reported.

Response:

We thank the Reviewer for identifying the insufficient methodological detail in the original submission. We revised the Methodology throughout, specifically in the *Data preparation*, *Models*, and *Validation strategies* sections.

Data preparation. We now specify the filtering sequence and parameters, the FastICA implementation, its EOG-proxy rejection criterion and random state, the common channel montage, and the 5-s windows with 0.5-s overlap. We also clarify that spectral power is estimated using Welch's method rather than STFT, with a Hamming window, 256-sample FFT and segments, and no overlap between Welch segments.

Models. We now report the fixed 168-dimensional classical feature representation (12 channels $\times$ 14 features per channel), the training-fold-only fitting of data-dependent transformations, XGBoost hyperparameters, the fixed random seed (42), and the absence of data-driven hyperparameter tuning.

Validation strategies. We now provide a fixed description of fold construction for every validation protocol, including the recording-unit grouping that keeps all overlapping windows on one side of a training-test split.

Comment 5

R1. The metric reporting is internally inconsistent. In single-label classification, support-weighted recall is mathematically equal to accuracy. Table 2 nevertheless reports, for example, 0.81 accuracy and 0.97 weighted recall for Full B, and 0.73 accuracy and 0.90 weighted recall for its cross-subject evaluation. Either the averaging description or the results are incorrect.

Response:

We thank the Reviewer for identifying this inconsistency. We conducted a full audit of the metric calculations and reporting and identified an error in the previously reported metric values. We corrected the error and recomputed the results for all experimental scenarios under the revised analysis workflow.

The revised Table 2 replaces accuracy and support-weighted recall with metrics that make class-specific performance and class imbalance explicit: balanced accuracy, macro F1, Matthews correlation coefficient (MCC), ROC-AUC, and recall for the low- and high-demand classes separately. The revised values are internally consistent; in particular, we no longer report support-weighted recall as a quantity distinct from accuracy in this single-label classification setting.

Comment 6

R2. Revise the presentation and interpretation. Figure 1 should include every validation family with uncertainty; Figure 2 should distinguish diagonal baselines from transfer cells.

Response:

We agree that the original presentation did not show the full evaluation design clearly enough. We have therefore added all validation strategies to Figure 3 in the Results section, together with uncertainty intervals for the displayed metric. This figure presents the within-dataset baseline, cross-subject, cross-session, cross-trial, cross-task and cross-dataset evaluations in one place.

[TODO — transfer-matrix presentation.] We also revised the transfer presentation so that within-composition baselines are distinguished clearly from source-target transfer results.

Comment 7

R3. The 15 references are relevant and include the original publications for both datasets, established cognitive-task literature and selected studies on EEG transfer. However, the coverage is too limited for the broad methodological claims advanced. Statements concerning DANN, ADDA, CORAL, TCA, Riemannian alignment, transformer architectures, self-supervised learning, XGBoost performance and low-density ICA are presented without appropriate citations.

Response:

We thank the Reviewer for this observation. We revised the literature framing and removed the unsupported catalogue of methods that were not evaluated in this study, including DANN, ADDA, CORAL, TCA, Riemannian alignment and self-supervised learning. We therefore no longer make claims about the performance of these methods in the present data.

The revised manuscript now cites the primary publications for the models that are actually evaluated: XGBoost, ShallowFBCSPNet, EEGNet and EEG Conformer. We also narrowed the discussion of the shared 12-channel montage and ICA to a limitation of the present analysis, rather than using it to attribute the observed transfer results to a particular source of signal variation.

Reviewer #2

Comment 1

Reconsider the target construct. The problem should be presented either as discrimination between operationally defined experimental conditions or as attention classification supported by independent attention measurements.

Response:

We adopted the first formulation. The revised manuscript defines the target as discrimination between protocol-defined higher-demand and lower-demand conditions, not measurement of an individual's level of attention. It therefore interprets the results as zero-shot transfer of this operational contrast and does not treat them as evidence for or against invariant neural markers of attention.

Comment 2

Sensitivity analyses should exclude drowsy or eyes-closed epochs and evaluate alternative label mappings.

Response:

We agree that drowsiness should not be silently merged with awake unfocused behaviour. The primary analysis now excludes the explicitly labelled drowsy segment. We cannot validly identify or exclude eyes-closed epochs from the available recordings: they contain no EOG channel, video, or epoch-level eye-state annotation. Frontal EEG channels were used only as proxies to identify EOG-like ICA components; they do not provide ground truth for eye closure. We therefore do not claim eyes-closed epoch rejection, because an unvalidated amplitude-based proxy could remove meaningful EEG activity.

[TODO — alternative label mappings.] The present revision does not add a formal sensitivity analysis over alternative label mappings. It reports the drowsy exclusion and the separate SAM-40 task compositions, but a full mapping-sensitivity analysis would require additional pre-specified analysis arms and is not yet part of the manuscript.

Comment 3

Reconstruct all validation splits at the participant, session, or trial level before window segmentation.

Response:

We reconstructed all validation protocols around recording-level metadata (participant, documented session, or trial) before assigning windows to folds. Every window from a recording unit remains on one side of a split. The revised Methods distinguish cross-session and cross-trial validation rather than using the terms interchangeably.

Comment 4

Every preprocessing transformation, including scaling and data-dependent feature processing, should be fit exclusively using the training data.

Response:

All learned transformations, including standardization, are contained in the estimator pipeline and are fit separately within each training fold. The handcrafted feature transforms are deterministic functions of an individual window and estimate no parameter from the data; they therefore have no fitting scope that could transmit information across folds.

Comment 5

Cross-task experiments should also use subject-disjoint source and target partitions.

Response:

We agree and rebuilt cross-task validation as leave-one-subject-out transfer. For each held-out participant, the source-task training partition excludes that participant and the target-task test partition contains only that participant. The source and target partitions are consequently subject-disjoint.

Comment 6

Audit and recompute all metrics from the saved predictions. Class counts, confusion matrices, per-class precision and recall, balanced accuracy, macro-F1, MCC, and AUROC should be reported alongside majority-class baselines.

Response:

We audited the reported metrics and recomputed them from the saved test predictions. The revised main results table reports balanced accuracy, macro F1, MCC, ROC–AUC and recall for each class; the analytical-composition table reports participants, recording units, window counts and class distributions.

[TODO — supplementary diagnostic outputs.] Confusion matrices, per-class precision and F1, and majority-class baselines are not yet presented in the manuscript or supporting information. They must be added before this response can claim that every requested diagnostic is reported.

Comment 7

Fold- or participant-level distributions, confidence intervals, and appropriate paired statistical comparisons are also required.

Response:

Figure 1 now displays uncertainty intervals for every validation family.

[TODO — distributions and paired comparisons.] The manuscript does not yet report fold- or participant-level distributions or paired statistical comparisons. These analyses must be added before this part of the request can be considered fully addressed.

Comment 8

Provide a complete and reproducible description of the methods, including filter characteristics, the ICA procedure, STFT settings, band-power calculations, feature dimensions, hyperparameters, tuning strategy, and random seeds.

Response:

We expanded the Methodology to specify filtering, ICA and the EOG-proxy criterion, channel alignment, window geometry, feature dimensionality, model families, random seed and fold construction. Hyperparameters were fixed before the benchmark comparison and each run stores its resolved configuration.

[TODO — complete parameter table.] The exact spectral-estimation settings and the full model hyperparameter list must be finalized in the supporting parameter table before resubmission.

Comment 9

At least simple linear and tree-based benchmark models should be included.

Response:

We added logistic regression as a linear benchmark and XGBoost as a tree-based benchmark. Both are evaluated under the same validation protocols as EEGNet, ShallowFBCSPNet and EEG Conformer.

Comment 10

An alignment or domain-adaptation baseline would further strengthen the transfer analysis.

Response:

We agree that an adaptation baseline would answer a complementary question. This revision, however, is deliberately limited to zero-shot, pre-adaptation benchmarks: no target data are used for fitting or calibration. We therefore state this boundary explicitly and do not present the zero-shot estimates as the expected performance of an adapted system.

[TODO — adaptation baseline.] No alignment or domain-adaptation method is currently implemented or evaluated; such a baseline must not be claimed in this response.

Comment 11

Revise the presentation and interpretation. Figure 1 should include every validation category with corresponding uncertainty estimates.

Response:

Figure 1 now includes all validation families and uncertainty intervals for the displayed balanced-accuracy estimates. It is generated directly from the recomputed analysis outputs.

Comment 12

Figure 2 should distinguish diagonal baseline results from transfer cells.

Response:

[**TODO — transfer matrix.**] A source–target transfer matrix has been generated in the analysis workspace, with matching within-composition baselines marked on the diagonal and directed transfer results outside it. It has not yet been incorporated into the manuscript as Figure 2; this response must remain provisional until that placement and numbering are finalized.

Comment 13

Table 2 should include sample sizes and corrected metric definitions.

Response:

The analytical-composition table now reports the number of participants, recording units, windows and the class distribution for every composition. The revised results table reports the corrected metric definitions and values.

[**TODO — final table numbering and supporting diagnostics.**] The final Table 2 layout and its relationship to the supporting diagnostic outputs remain to be finalized before submission.

Comment 14

Claims regarding hardware, causal language, references, and the secondary data-use ethics statement should also be corrected.

Response:

We corrected the description of the acquisition setup: the datasets share the sampling rate and voltage resolution but not an identical electrode montage. We also narrowed causal and physiological interpretations, revised the bibliography as described in our response to Reviewer 1, and describe the work as a secondary analysis of public, de-identified datasets.

[**TODO — ethics statement.**] The exact secondary-data-use ethics statement requires confirmation from the authors and must be inserted in the manuscript before resubmission.

Reviewer #3

Comment 1

I have carefully reviewed the article. At a first glance, this paper would have been really very interesting but unfortunately it is really too short and superficial for the promises it makes at the beginning. The authors set up a compelling premise in the introduction, promising to systematically evaluate how neural markers of attention transfer across diverse paradigms and validation strategies. However, the execution falls completely flat. It would be necessary to compare all the most advanced deep models for EEG decoding at the state of the art and see if, by passing from one task to another, they do better or worse. Instead, here unfortunately there are very few discussions on state of the art models and on datasets used for many tasks such as motor imagery, sleep, and other cognitive states. The entire methodology relies on a simplistic feature extraction pipeline combined with a traditional machine learning classifier, which completely ignores the massive advancements made in deep representation learning for electroencephalography over the last several years.

Response:

We appreciate this concern. The revised benchmark evaluates two classical baselines (logistic regression and XGBoost) alongside three EEG-specific deep architectures: ShallowFBCSPNet, EEGNet and EEG Conformer. The deep models learn from prepared channel-by-time EEG windows and are evaluated under the same validation protocols as the classical models. We have also narrowed the claims to the two available datasets and their protocol-defined conditions; the study is not presented as an exhaustive survey of EEG decoding.

[TODO — broader literature context.] The Introduction still needs a concise, cited discussion that situates these three architectures within the broader EEG deep-learning literature.

Comment 2

When discussing the state of the art in electroencephalography decoding, it is entirely unacceptable to limit the analysis to gradient boosted decision trees like XGBoost fed with basic short time Fourier transform spectral power features. The field has moved well beyond these classical baselines. Advanced deep models such as convolutional neural networks specifically designed for time series, spatial temporal graph convolutional networks, and transformer based architectures have become the standard for brain computer interfaces and cognitive state monitoring. The manuscript barely touches upon these architectures, briefly mentioning transformers in the related work but failing to implement or benchmark any of them. To truly understand whether neural signatures of attention are invariant across paradigms, one must leverage models capable of learning hierarchical spatial and temporal representations directly from the raw or minimally preprocessed signals. By relying on predefined spectral bands and overlapping windows, the authors impose a rigid inductive bias that likely destroys the subtle, task invariant dynamics they claim to be looking for. Therefore, the conclusion that current classifiers capture context dependent features rather than invariant signatures is heavily confounded by the fact that the authors used a very rudimentary classifier that lacks the capacity to learn invariant representations in the first place.

Response:

We added EEGNet, ShallowFBCSPNet and EEG Conformer as end-to-end benchmarks on prepared channel-by-time EEG windows, alongside the feature-based logistic regression and XGBoost baselines. This design tests whether the pattern across validation protocols persists when the decoder learns its representation from the signal rather than relying only on predefined spectral features. We do not, however, interpret any single model's result as proving or disproving the existence of invariant attention representations.

Comment 3

Furthermore, the narrow focus on a passive control task and short cognitive tests like the Stroop test and mental arithmetic severely limits the scope of the claims regarding cross task generalization. A comprehensive evaluation should have included datasets spanning a much wider variety of neurological tasks. For instance, motor imagery is one of the most widely studied paradigms in the field, and analyzing how attentional engagement during motor imagery transfers to active cognitive tests or sleep staging would have provided a much stronger foundation for the paper. Sleep datasets, in particular, offer a continuous spectrum of arousal and attentional states that could have perfectly complemented the drowsy condition in the first dataset. The lack of discussion on these broader applications and the datasets commonly used for them makes the paper feel isolated from the broader brain computer interface community and leaves the reader wondering how these models would perform in real world clinical scenarios.

Response:

We agree that the two datasets limit the scope of the benchmark. We did not add motor-imagery or sleep-staging datasets because their labels define different target constructs from the protocol-defined high-demand versus low-demand contrast evaluated here. Adding them would require a separate harmonization and validation rationale and would not, by itself, establish transferability of the current target contrast.

[TODO — scope statement.] The manuscript should state this boundary explicitly and discuss the limited generalisability beyond the two included public datasets before submission.

Comment 4

To rectify these significant omissions, the authors must update their literature review and theoretical framework by citing and discussing recent works that successfully apply deep learning to complex electroencephalography tasks. I strongly require the authors to cite the paper titled *Investigating the impact of rational dilated wavelet transform on motor imagery EEG decoding with deep learning models* published in the year 2025. It is extremely important to cite this work because it clearly demonstrates how advanced time frequency transforms can be integrated with deep learning models to significantly improve decoding performance on complex tasks like motor imagery. This highlights the inadequacy of the standard short time Fourier transform approach used in the current manuscript. Additionally, the authors must cite the paper titled *RatioWaveNet: A Learnable RDWT Front-End for Robust and Interpretable EEG Motor-Imagery Classification*, also published in 2025. Citing this paper is crucial because it showcases the current state of the art in developing robust, interpretable front ends that are completely learnable, effectively replacing the static feature extraction pipelines that the authors currently rely on. Since these papers are very recent, they represent the cutting edge of the field and provide exactly the kind of state of the art methodological context that this superficial manuscript currently lacks.

Response:

We reviewed the two recommended papers. Both concern motor-imagery decoding, which is outside the empirical scope of this study; they cannot support claims about the performance of the present models or datasets.

[TODO — citation decision.] If the revised Introduction retains a broader discussion of deep EEG decoding, we will cite these papers only as clearly delimited motor-imagery examples. They are not yet cited in the manuscript and must not be represented as evidence for the present operational contrast.

Comment 5

Moving to the semantic and conceptual issues within the manuscript, there are several major flaws that undermine the validity of the unified labeling scheme. The authors decided to merge the drowsy state, the unfocused state, and the resting state into a single low attention class. This is a severe semantic error. From a neurophysiological perspective, drowsiness involves a transition into early stage sleep, characterized by the emergence of theta waves, vertex sharp waves, and a general withdrawal of environmental awareness. This is fundamentally different from an awake but unfocused state, which might involve default mode network activation and mind wandering, or a resting state with eyes open or closed, which is typically dominated by posterior alpha oscillations. By grouping these distinct physiological phenomena into a single binary class, the authors create an artificially heterogeneous category that confuses the classifier. This semantic misalignment invalidates many of the cross dataset conclusions, as the model is likely learning to distinguish arousal levels rather than attention levels.

Response:

We accept this criticism. Drowsiness is not treated as interchangeable with an awake unfocused state or rest. The explicitly labelled drowsy phase is excluded from window generation and classifier evaluation; the retained lower-demand conditions are Unfocused and Rest. We also narrowed the interpretation of all results to protocol-defined conditions and do not claim that the model isolates attention from arousal, ocular activity or other physiological differences.

[TODO — label terminology.] The table heading and class labels should be revised from “attention” to protocol-defined demand terminology before submission.

Comment 6

Furthermore, framing mental arithmetic purely as an attention task ignores the massive working memory load it requires, which produces entirely different cortical activation patterns compared to the simple visual monitoring required in the train simulator task.

Response:

We agree. Mental arithmetic is not described as a pure attention task in the revision; its working-memory demand is one component of the protocol-defined higher-demand condition. This is why the manuscript does not infer a single cognitive mechanism from successful or unsuccessful transfer between task compositions.

Comment 7

Similarly, the grammatical and syntactic quality of the manuscript leaves much to be desired. There are numerous instances of poor phrasing and missing punctuation that disrupt the reading flow and obscure the scientific meaning. For example, in the description of Dataset A, the authors write that all experiments were performed with volunteer participants chosen among students. This phrasing is awkward and unscientific; it would be much better to state that the participant cohort consisted of university student volunteers. In the description of Dataset B, there are glaring punctuation errors in the descriptions of the tasks. The text reads Stroop Color Word Test participants were required to identify the color, and Arithmetic problem solving participants mentally solved a mathematical problem, and Mirror Image Recognition participants were shown images. In all these cases, there is a missing colon or em dash after the name of the task, making it seem like the participants themselves are named after the tasks. These are careless grammatical errors that should have been caught during proofreading. Furthermore, the phrasing in the results section, specifically the sentence stating that accuracy and F1 score reaching 0.86 for Dataset A and 0.81 and 0.88 respectively for Dataset B, is disjointed and confusingly structured. The authors also failed to properly format the cross dataset directions in their text, sometimes omitting the necessary words to explain the direction of the transfer.

Response:

We carefully proofread the revised manuscript and corrected the examples noted by the Reviewer, including the participant description, punctuation in the task descriptions and wording of directed source–target transfers. The earlier disjointed sentence reporting accuracy and F1 has been replaced by the revised results presentation.

[TODO — final language edit.] The full manuscript requires a final language edit after the remaining Results, Discussion and Abstract text has been finalized.

Comment 8

Another critical methodological failure that must be addressed is the cross task validation experimental design. The authors openly admit in the limitations section that in their cross task experiments, the source and target tasks were drawn from the same pool of Dataset B participants, and consequently, task transfer and subject identity are not fully disentangled. This is not just a minor limitation; it is a fatal flaw for a study whose entire premise is about evaluating transferability and generalization. If the model is evaluated on a new task but with the same subjects it was trained on, any successful transfer could merely be the result of the classifier learning subject specific noise profiles, baseline anatomical features, or electrode impedance artifacts, rather than genuine task invariant attention markers. A true cross task evaluation must be performed using a leave one subject out approach across tasks, ensuring that the model is tested on both unseen tasks and unseen subjects simultaneously, or at the very least, evaluated on completely disjoint subsets of subjects.

Response:

We agree and rebuilt the cross-task protocol as leave-one-subject-out transfer. In each fold, source-task training data exclude the held-out participant and target-task testing uses only that participant. The revised design therefore tests both an

unseen task composition and an unseen participant.

Comment 9

The authors also acknowledge that their within dataset baselines are likely overly optimistic due to the use of densely overlapping windows. They state that overlapping one second windows with a zero point one two five second step size lead to highly autocorrelated samples, and that splitting these windows randomly across cross validation folds causes temporal leakage. While it is good that they acknowledge this, merely stating it in the limitations is insufficient for a paper focused on validation strategies. The core contribution of this paper is supposed to be a rigorous evaluation of validation schemes, yet they purposefully include a flawed baseline that they know is leaking data. They should have implemented the group aware splitting by trial or session that they suggest as future work directly in this manuscript. Presenting inflated within dataset metrics undermines the entire comparative analysis, as the drop in performance observed in cross subject and cross session scenarios might just be the result of removing data leakage rather than actual domain shift. The failure to correct this known issue makes the baseline metrics virtually meaningless.

Response:

We agree and removed this leakage from the baseline. Windows are now 5 s with 0.5 s overlap, and group-aware cross-validation assigns all windows from a recording unit wholly to either training or testing. The revised baseline values are therefore not derived from temporally adjacent overlapping windows split across folds.

Comment 10

Additionally, the reliance on a reduced twelve channel montage significantly limits the potential of the study. While aligning the spatial resolution of the two datasets is a logical step for cross dataset evaluation, dropping from thirty two channels down to twelve discards a massive amount of spatial information that could be vital for deep learning models to learn invariant representations. The authors note that this low density configuration limits the stability of the independent component analysis used for artifact removal, which introduces yet another confounding variable. If the artifacts were not cleanly removed due to the lack of spatial channels, the classifier might simply be tracking eye movements, blinks, or muscle tension rather than cortical attention networks. This is especially problematic given that the Stroop test and mental arithmetic often elicit different physical responses, facial microexpressions, and stress induced muscle artifacts compared to a passive train driving simulator. The failure to adequately separate these physiological artifacts from true neural signals casts doubt on whether the model is actually decoding attention at all.

Response:

We clarify that ICA is fitted and applied using all EEG channels available in each original recording unit; the common 12-channel montage is selected only after artifact cleaning, before windowing and model fitting. The shared montage therefore standardizes the model input for cross-dataset comparison rather than reducing the dimensionality of the ICA decomposition.

We nevertheless acknowledge a residual limitation: neither dataset provides an independent EOG channel, so frontal EEG channels serve only as EOG proxies. This does not provide ground-truth validation of ocular-artifact removal, and our interpretation does not attribute model decisions to cortical attention signals.

Comment 11

Finally, the absence of any discussion regarding model explainability or feature importance is a missed opportunity. Even with a simpler model like XGBoost, the authors could have analyzed which spectral bands and channels were most critical for the cross task and cross dataset predictions. Without an analysis of the underlying features driving the decisions, the claim that models capture context dependent features remains purely speculative. State of the art deep models, such as the ones recommended for citation earlier, often include interpretability modules or attention mechanisms that allow researchers to visualize the spatial and temporal features the network relies on. By neglecting this aspect, the authors fail to provide any neuroscientific insights into why the transfers failed, leaving the paper as a purely empirical exercise in applied machine learning with poor experimental controls.

Response:

[TODO — feature attribution.] Per-fold coefficients for logistic regression and feature importances for XGBoost are saved with the experiment outputs, but they have not yet been aggregated or reported in the manuscript. We will not claim an explainability analysis until these descriptive model-specific results are presented; they would not explain the decisions of the

deep models.

Comment 12

In conclusion, the manuscript fails to deliver on its ambitious initial premises. To be considered for publication, it requires a complete overhaul. The authors must replace their obsolete feature extraction and gradient boosting pipeline with a suite of advanced deep learning architectures, comparing their performance across various transfer scenarios. They must expand their dataset inclusion to encompass a wider variety of cognitive states and tasks, particularly motor imagery and sleep stages, to provide a true measure of paradigm invariance. They must correct the severe semantic errors in their label harmonization process, particularly the grouping of sleep related states with resting states, and fix all grammatical mistakes and punctuation omissions throughout the text. Most importantly, they must implement rigorous, leakage free validation schemes for all their baselines and ensure that cross task evaluations are completely separated from subject identity confounds. Without these major revisions and a significant expansion in scope, the paper remains too superficial and methodologically flawed to contribute meaningfully to the state of the art in electroencephalography decoding.

Response:

We appreciate the detailed assessment. The revision replaces the leaking baseline and the subject-confounded cross-task protocol, excludes the drowsy phase from classifier evaluation, adds linear and tree-based benchmarks alongside three EEG-specific deep architectures, and narrows the interpretation to protocol-defined conditions. The remaining limitations are stated explicitly in the point-by-point responses above.

[TODO — final consistency audit.] Before submission, the remaining manuscript TODOs, supporting diagnostics, transfer-matrix placement, ethics statement and final language edit must be completed and reconciled with this letter.